Simulating One DOF Attitude Control Using a Reaction Wheel

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I. Introduction

Reaction wheels are a critical component in spacecraft attitude control systems by allowing for precise orientation without fuel usage. They operate using principles of conservation of momentum to generate rotation by generating angular momentum in the opposite direction of necessary movement.

This project aims to design and construct a mock reaction wheel assembly to demonstrate and test these principles in a controlled environment. The system uses a free-spinning test platform to act as a spacecraft and contain all necessary hardware components. A motor drives a high-inertia reaction wheel to generate angular momentum, causing the motor to spin in the opposite direction. The system has tracking and self-stabilization software to mimic programs used in spacecraft control.

This approach provides a simple and scalable method for studying spacecraft attitude control while strengthening understanding of robotics and aerospace concepts. The system's design also works as an educational tool to explore dynamics, circuitry, embedded systems, and feedback control loops.

II. Methodology

A. Mechanical Design

In order to create an effective solution that could simulate self-stabilizing attitude control, the design had to meet certain criteria: a large moment of inertia to create necessary counter-momentum, a base that would prevent tipping due to slight weight imbalances, and a way to integrate both while allowing the testbed to spin freely. The inertia was generated by a reaction wheel that was spun by a motor on the testbed. The base was wide enough to prevent tipping unless at extreme angles and included support for the shaft that connected to a provided test bed. A bearing block fully supported and allowed for free spinning of the test bed.

The flywheel inertia had to be maximized so that the least possible amount of energy would be used. The principle of conservation of angular momentum and the formula for angular acceleration were utilized to find an inertia that would allow for a motor saturation time to be greater than 10 seconds. The REV Robotics HEX motor utilized had a maximum rotational velocity of 1500 RPM and maximum torque of .105 $N \cdot m$. To account for air resistance, the saturation time used in calculations was 35 seconds. The angular acceleration was 4.488 rad/s^2 and the minimum inertia was 0.0234 $kg \cdot m^2$. Due to the lightweight and unpredictable mass properties of PETG filament, 12 1/2-20 bolts and nuts with a combined mass of 1.05 kg were used as the sole method of inertia calculation. The minimum radius was calculated as 0.150 meters, and the flywheel was designed as such. It was made in three parts to fit a standard Bamboo 3-D printer with slats that would screw on to connect each part. Spacers were made for each screw to prevent slide.

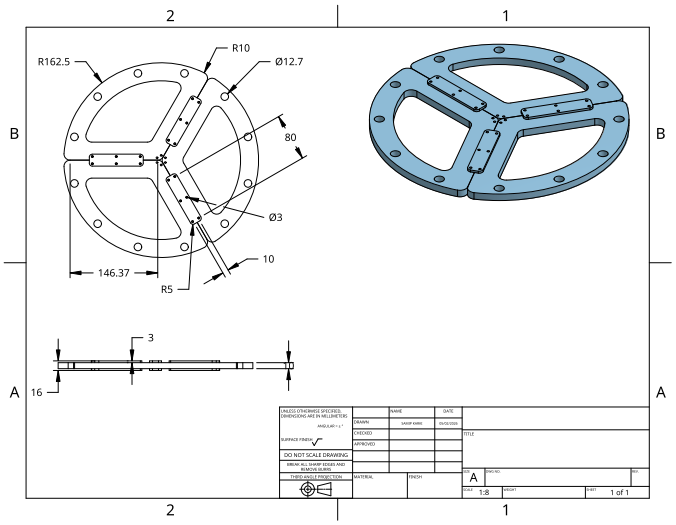


Fig. 1. Reaction Wheel

The base was designed to prevent the reaction wheel assembly from tipping due to weight imbalances during spinning and to support the shaft that would integrate with the bearing block. To enact tipping angle analysis, the mass of the entire reaction wheel system was found through a scale, which came to 2.431 kilograms. Then, the center of masses were taken for the most mass intensive parts, those being the testbed platform, base, and reaction wheel. The centroid of the entire system was found by adding the centroids of the mass intensive parts and dividing them by the total mass. Knowing that the wheel will tip from the rim, the tipping angle was found using the formula: $\theta = \arctan\left(\frac{r-d}{h}\right)$, where h is the height from the base to the y-axis centroid of the system, r is the radius of the base, and d is the horizontal distance from the rim to the combined x-axis and z-axis centroid of the system. Through this, the tipping angle was found to be 46.94°. Using an inclinometer, the actual tipping angle was found to be an average of 24.27°.

The shaft constraint was at the center of the base. The

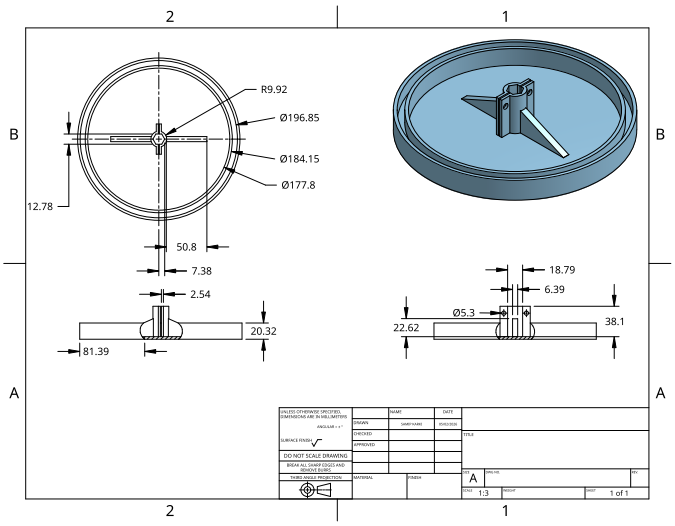


Fig. 2. Base

constraint was made of 2 separate walls that were each a half-hexagon shape with extensions on the sides where screws could be screwed in to tighten the constraint around the shaft and prevent it from sliding out. Each wall had a chamfer on the outside that would support the shaft further.

The bearing block had to fully constrain the non-rotational movement of the shaft in a way that would still allow for free-spinning of the test bed. The bearing block was designed to have a through hole larger than the shaft to prevent friction. It had a platform on the top where the test bed would screw on that would not interfere with other test bed components. On the top and bottom of the bearing block there was a slot for bearings to flush fit. Both bearings were necessary to fully constrain vertical movement of the shaft.

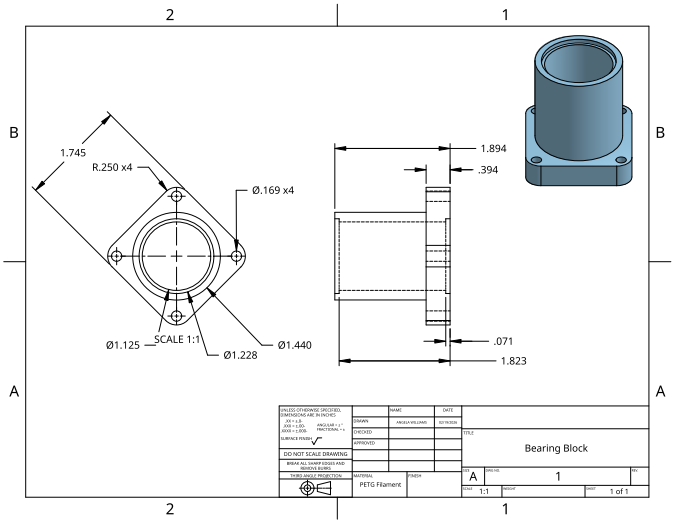


Fig. 3. Bearing Block

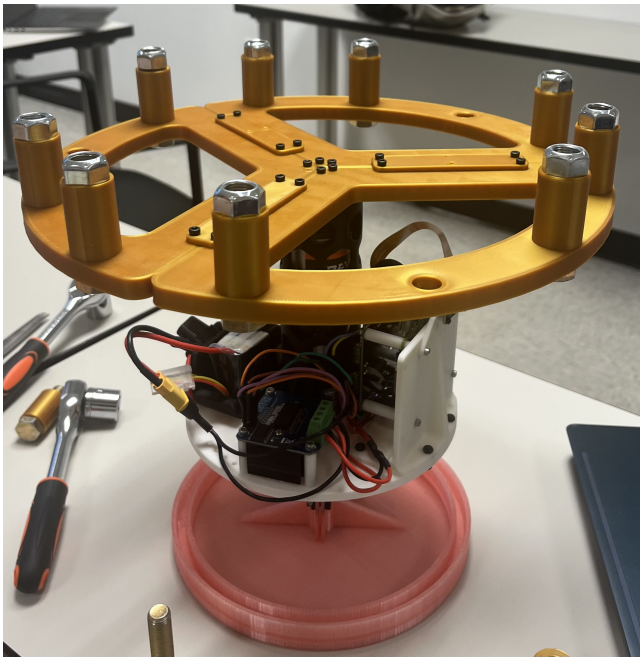


Fig. 4. Fully Assembled Reaction Wheel System

B. Electrical Integration

The hardware required multiple electronics to power and control the wheel including a battery, fuse, switch, Raspberry Pi 5, a Raspberry Pi Power Board, Adafruit 9-DOF IMU, REV Robotics HEX motor, and an H-bridge motor controller.

An 11.1 volt 3-cell Lithium Polymer battery pack was used to power the reaction wheel assembly. The battery was connected to a fuse to prevent over current on critical electrical components, which was also connected to a switch to turn on and off the system as wheel as act as an emergency stop. The wire leading from the switch and the other wire leading from the battery each split into the RPi 5 Power Board and H-Bridge. The RPi 5 Power Board was linked to the RPi 5 through a USB-C wire, which was connected to the IMU, motor, and motor controller. A camera was mounted into the RPi 5 using a ribbon cable. The wiring diagram is shown in Figure 5.

The IMU collected 9 degrees of information and communicated with the RPi 5 via I^2C, which allows bidirectional signal transference using the SDA and SCL slots on the IMU. The IMU was powered through Vin and GND pins on the RPi 5. The RPi 5 communicated with the H-Bridge to control forward motion, backward motion, braking, and speed of the motor by connecting to the R_EN, L_EN, RPWM, and LPWM pins on the H-bridge. The H-Bridge was connected to the HEX Motor via B+ and B- connections to control current to the motor. To obtain motor data, the encoder Channel A, Channel B, 3.3V, and GND ports were connected to the RPi 5.

The wires utilized were 18 gauge and 24 gauge, with 18 gauge wires being used for more voltage intensive connections, and 24 gauge for the remainder of the lower power connections

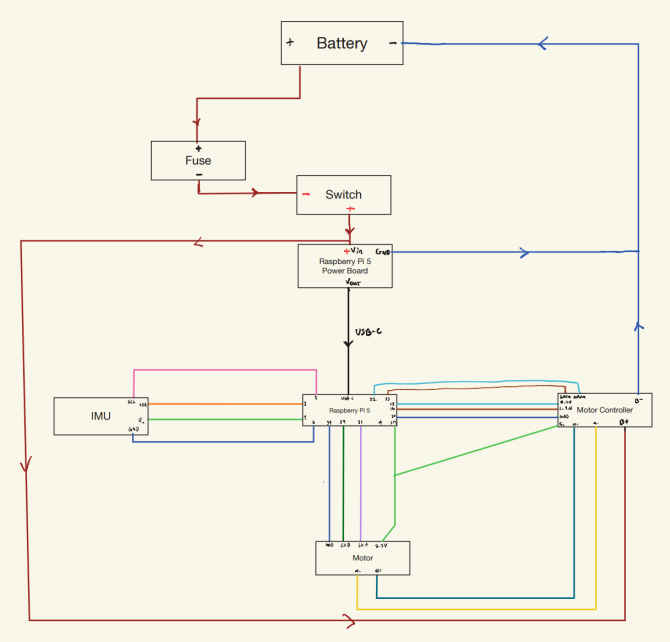


Fig. 5. A wiring diagram to plan connections

which mostly carried signals. The necessary lengths of the wires were measured while the test bed components were fully assembled and then cut 4 centimeters longer than measured to account for error. The wire splits, battery connections, and IMU pins were then soldered as shown in the connection chart in Figure 6. A physical representation of the connections are shown in Figure 7.

Wire Chart				
Labels	**Color**	**Length**	**Connections**	**Gauges**
A	Red	6cm	Battery -> Fuse	*18 Gauge*
B	Red	13cm	Fuse -> Switch	*18 Gauge*
C	Red	24 cm	Switch -> Power Board	*18 Gauge*
C	Red	26 cm	Switch -> Motor Controller	*18 Gauge*
S	Black	17 cm	Battery -> B- MoCo	*18 Gauge*
T	Black	17 cm	Power Board SPLIT -> Motor Controller	*18 Gauge*
R	Black	22cm	M- Motor Controller SPLIT-> M- Motor	*18 Gauge*
Q	Red	22cm	M+ MoCo -> M+ Motor	*18 Gauge*
O	Brown	15 cm	IMU GND -> Pin 6 (Pi GND)	24 Gauge
N	Green	15cm	IMU Vin -> Pin 1 (3.3V Pi)	24 Gauge
M	Blue	15 cm	IMU SCL -> Pin 5 (GPIO 3)	24 Gauge
L	Yellow	15 cm	IMU SDA -> Pin 3 (GPIO 2)	24 Gauge
U	Purple	22 cm	Pin 32 -> RPWM	24 Gauge
I	Purple	22 cm	Pin 15 -> R_EN	24 Gauge
V	Orange	22 cm	Pin 33 -> LPWM	24 Gauge
J	Orange	22 cm	Pin 16 -> L_EN	24 Gauge
K	Brown	22 cm	Pin 20 -> GND MoCo	*18 Gauge*
P	Green	22 cm	Pin 17 (3V power Pi) SPLIT -> Vcc MoCo	24 Gauge
G	Green	20 cm	Pin 17 SPLIT -> 3.3V Motor	24 Gauge
E	Blue	20 cm	Pin 31 -> CH A Motor	24 Gauge
F	Yellow	20 cm	Pin 29 -> CH B Motor	24 Gauge
H	Brown	20 cm	Pin 34 -> Motor GND	24 Gauge

Fig. 6. A chart of wire connections in the system

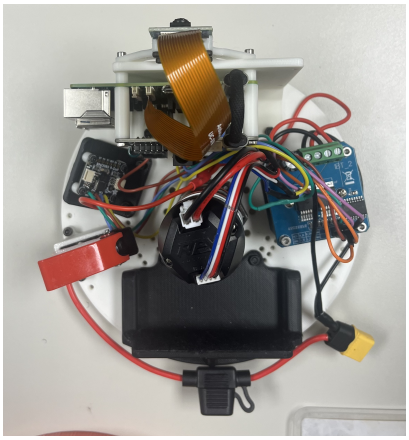


Fig. 7. The physical wiring set-up

C. Testing

The three objectives to this project were to return the test-bed rotational velocity to zero after disturbance, to track an object of predetermined color, and prevent motor saturation for over ten seconds. Returning the testbed to rest required the flywheel to spin in the same direction of the induced speed to create a counter-moment. The information necessary to complete this objective was the rotational speed of the platform, which was pulled from the IMU and a PID controller, and the fly wheel speed, which was pulled by the motor encoder. Based on the speed that the IMU read, the motor would spin at a certain duty in the same direction to bring the system back to rest. For example, if the testbed was spun clockwise, the flywheel would also spin clockwise, thus inducing counterclockwise moment on the testbed which counteracted the applied rotation. To avoid noise affecting the readings, a dead-band of about ±0.007 rad/s was set in place. A PID controller was created to control the reaction of the flywheel. To tune it, the Kp was increased until the oscillations that were created by disturbances increased over time. Then, the Kd was increased until the oscillations were not noticeable. Last, Ki was increased until the drift over time stopped.

These ideas were then carried over and applied to achieving the second goal. Using OpenCV, specifically cv2, the color chosen was encoded using HSV (Hue, Saturation, Value) rather than RGB (Red, Green, Blue) values due to its increased performance over varying light conditions. The system was designed to switch between 2 states: SEARCH and LOCKED. If the code was in SEARCH mode, then the platform would spin slowly in the direction the object was last seen until it found the object, where it would then switch to LOCKED mode and stop spinning. If it was on its first search cycle after startup, the code would default to spinning clockwise. Once the object was found, the camera would “lock on” to the object, following it around using a PID control based on the offset of the object from the center of the frame. This allowed the object to be moven at largely varying speeds and for the camera to accurately track it until the object dropped

out of view. When the object dropped out of view for more than 20 frames, the system would switch back to SEARCH mode.

The third and final goal, as mentioned before, was to ensure that the wheel did not saturate, or reach its max speed, in under 30 seconds. To do this, code was generated to spin the wheel at a safe but high duty cycle for 30 seconds while outputting RPM values. If the wheel did not reach max speed at any time, it would have succeeded in the saturation test.

III. Results

A. Goals

For goal 1, the wheel was successful in responding quickly to physical disturbances with a slight drift. With most disturbances, however, the wheel spun down quickly enough to where the testbed stayed motionless and the software was reading 0 RPM in both the platform and wheel.

For goal 2, the wheel was again successful in stopping in response to seeing the color green. With even subtle appearances, the wheel successfully oriented the testbed to face the color with a quick stop. Tracking the color as it moved slowly from either left or right also proved successful, with the camera following smoothly at a slow rate. When not in view of the color green, the wheel spun smoothly as it searched for the color again. A limitation of this design was that the wheel did not move smoothly when faced with green at a distance. The perturbations in motion caused the wheel to either ignore the color completely, or overshoot its spin and create a cycle of oscillating until it finally slowed and stopped at the color.

For goal 3, the wheel was successful in not reaching saturation before 10 seconds. For three trials, the speeds reached at the goal time are shown below in Figure 8, with the saturation speed being 6000 RPM.

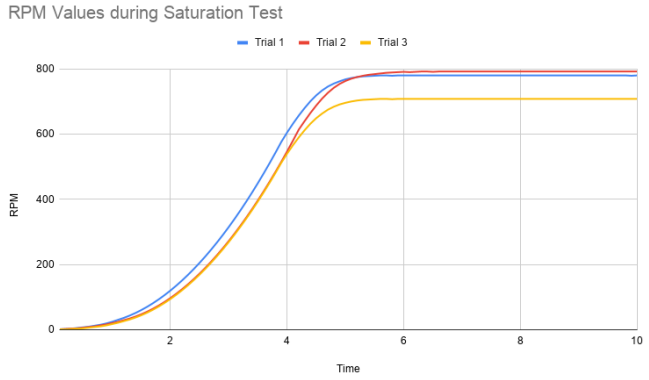


Fig. 8. RPM Values during Saturation Test

B. Data Collection

The PID control and angular velocity responses of the platform speed were collected by taking csv (comma separated values) data in a file from demonstrations and importing them into a graph.

For goal 1, in Figure 9's graph shows that the Proportional term in PID contributes the most to disturbances, which are shown in the blue lines spiking. A total disturbance cycle (disturbance to rest), is shown in the red box. A notable piece of data is that the average settle time is very low, which means that the wheel returns to rest in a very small amount of time, most likely due to the powerful P term fighting back against the disturbance. However, the stable %, which denotes the percentage of the total time that the platform actually stays stable after returning to 0 RPM, is very low. The initial red boxes depict slight disturbances in the beginning of testing, and the disturbances without a red box show oscillatory motion as the wheel attempted to go to rest.

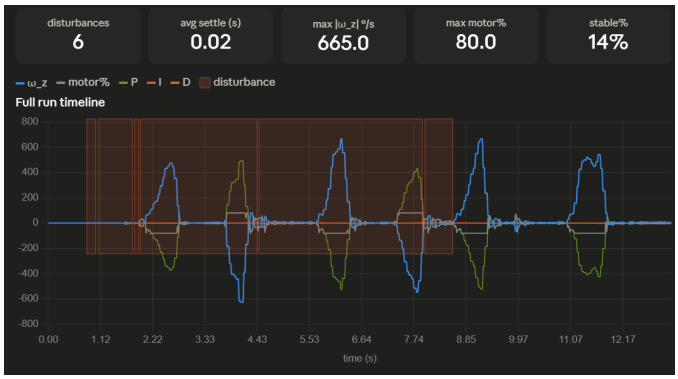


Fig. 9. PID control for Goal 1
Disclaimer: Claude generated graph using csv data during testing

For goal 2, Figure 10's graph depicts the wheel moving between the locked and search state when attempting to detect the color green. When the camera locks onto the color, the PID attempts to drive offset to 0 (-1 means the camera is shifted fully left, while 1 means the camera is shifted fully right). To do this, the P term is used while I, the integral term, attempts to correct drift and D, the derivative term, prevents overshoot. The graph shows that as the offset grows smaller, the D term and motor spike to brake the motor and lower the P term's overshooting when it first encounters the color green. In the graph below, the orange blocks depict a searching state while the green and yellow blocks depict a locked state. The pink oscillating lines depict the platform's rotation rate, which runs until the camera fully locks onto the color when the system rests.

IV. Future Work

A. Limitations and Improvements

The slight drift of the testbed platform in goal 1 was speculated to be due to the wheel's properties of mass far exceeding that of the testbed. Lingering inertia drew the testbed to start moving in the direction of the wheel's spin if the wheel did not stop quickly enough, which was depicted in the PID graph through the low stable % term. This could be remedied with further PID tuning to ensure the wheel would oscillate less. Another potential remedy could be applying



Fig. 10. PID Control for Goal 2
Disclaimer: Claude generated graph using csv data during testing

code that applies a large counter-spin to the wheel, causing the wheel to brake in place sooner.

For goal 2, it is thought that the rotational speed of the wheel as well as its inertial properties due to its large mass in comparison to the testbed would cause it to often pass by the color green long enough to move onto the next search cycle. This is also shown in Figure 10, where the D term in PID control spikes up as the wheel spins faster to stop the camera from drifting. However, too large of a spike in the D term denotes more overshoot or oscillatory behavior. When in the next search cycle not detecting the color green, the wheel would forget that it had seen the green and continue to search for it. If in the same search cycle, the wheel would start oscillating towards the position where the camera had previously detected the color as it slowly spun down to rest. To fix these issues, the camera can be upgraded for a better frame rate. The original frame rate of the camera used was 30 Hz, which was slower than the rate detection was calculated, as the main loop runs at 50 Hz. Cutting the resolution by 4 increased the frame rate by the same factor, allowing quicker detection and smoother following, but this was done at the cost of resolution, limiting the distance the camera could detect from. Upgrading the camera to one that has a higher frame rate without sacrificing resolution would allow for a larger range of detection and for smoother following.

B. Further Testing and Application

To implement and ensure that the implementations would strengthen the functionality of the wheel, some new tests can be applied. For one, a distance test can be implemented to test how accurate and quickly the wheel can detect objects at a further than usual distance. This can be done by using a camera with a higher frame rate and with improvements to the PID control and motor speed. Another test could also involve object tracking based on light to simulate real space-based telescopes which track stars. Once this system is refined, real world applications could include use in spacecraft orientation and tracking systems. The software developed to complete the first goal, for example could be used to set a desired position and spin a spacecraft to that position. It could also be used to reorient itself in case of unwanted movement. The program developed for goal 2 could be used in the tracking of celestial objects by terrestrial and extraterrestrial telescopes by locking on to a desired object and moving the telescope to track it.

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